

Transiberia — EU AI Act Risk Assessment

Scope: AI capabilities in the n8n on-premise system · All processing local · No external data sharing · September 2026

Overall verdict: Minimal to Limited Risk — no high-risk AI in scope. The on-premise architecture keeps all data within Transiberia's own server, eliminates any third-party AI provider dependency, and makes explainability obligations straightforward to satisfy. All compliance obligations are manageable within Phase 0 of the engagement.

1. AI CAPABILITY INVENTORY & RISK CLASSIFICATION

AI capability	What it does in plain terms	Data processed	Risk tier	Key obligation
Document & message reader (local LLM)	Reads carrier WhatsApp messages and PDF documents. Extracts structured fields: load reference, route, carrier, status, rate, ETA. Runs entirely on-premise — no data leaves Transiberia's server.	Carrier names, contact details, shipment data, delivery notes. Processed locally. Never sent to any external service.	MINIMAL RISK	Human reviews extracted output before any action. On-premise processing satisfies GDPR data residency.
Exception classifier (local LLM)	Reads incoming updates and GPS signals, assigns a status: on track / likely late / confirmed late / missing / needs review. Outputs a confidence score alongside the classification.	Shipment status messages, GPS position, carrier history from on-premise database.	MINIMAL RISK	Classification is advisory. Dispatcher confirms before any escalation or reroute action is taken.
Route deviation detector	Compares live GPS position against the pre-calculated expected route and schedule stored at load creation. Flags deviations beyond configurable thresholds of distance or time.	GPS coordinates every 5 minutes, pre-calculated route baseline, ETA — all stored in on-premise database.	MINIMAL RISK	Alert is informational. No automated action taken without dispatcher confirmation. Thresholds set by Transiberia ops team.
Carrier scorer (SQL + ML ranking)	Scores and ranks available carriers for a new load based on historical on-time rate, exception rate, and lane performance. Outputs a ranked shortlist for dispatcher review — not an automated award.	Historical load outcomes and carrier performance data sourced entirely from Transiberia's own on-premise database. No external data.	MINIMAL RISK	Dispatcher always makes the final carrier selection. AI provides ranked options, not a binding recommendation.
WhatsApp automation (bot)	Sends automated booking confirmations, delay alerts, and proof-of-delivery requests to carriers via WhatsApp Business API. Receives and parses carrier replies.	Carrier phone numbers, shipment reference, status updates.	LIMITED RISK	Carriers must be informed they are interacting with an automated system. Disclosure in onboarding documentation and in the opening message of every automated thread.
Predictive exception detection (Phase 2)	Future capability: flags loads likely to generate exceptions before any problem is reported, based on patterns in historical data. Not in Phase 1 scope.	Historical load and exception patterns, carrier reliability scores — all on-premise.	MINIMAL RISK	Output is advisory — an early warning signal only. Human dispatcher decides whether and how to act.

Why no high-risk classification? EU AI Act Annex III defines high-risk as AI making decisions in areas such as critical infrastructure safety, employment, credit scoring, law enforcement, or essential public services. Transiberia's AI assists internal logistics operations — it does not affect people's rights or access to services. **Review required if** Transiberia later introduces AI-driven carrier credit scoring, automated driver performance evaluation, or any AI affecting individual employment decisions.

2. THE SIX COMPONENTS OF THE ASSESSMENT

1. AI system inventory

A register of every AI capability in use or planned — what it does, what data it processes, what decisions it influences, and which model version is running. Updated whenever a capability is added or materially changed.

2. Risk classification

Each capability classified against the four EU AI Act tiers with documented reasoning. Because all processing is on-premise and no capability affects people's rights or access to services, all current systems are Minimal or Limited Risk.

3. Transparency — carrier disclosure

The WhatsApp automation bot (Limited Risk) requires carriers to be informed they are communicating with an automated system. A short, clear disclosure is added to carrier onboarding documentation and to the opening message of every automated thread.

4. Human oversight — by design

Every AI output in this system is advisory. No automated action is taken without a human confirmation step. This is designed into the workflow, not bolted on as policy. The dispatcher confirms carrier selection, escalation decisions, and reroute actions.

5. Data governance

All data processed by the AI capabilities stays on Transiberia's own server. No data is sent to any external AI service or cloud platform. GDPR obligations: carrier contact data retention limits, purpose limitation documentation, and access controls on the local server.

6. Ongoing monitoring

Quarterly review of AI accuracy: exception classification error rate, carrier scorer recommendation acceptance rate, route deviation false-positive rate. Results logged in the on-premise database. Assessment revisited whenever a new AI capability is added or the local model version is updated.

3. RECOMMENDATION & COST

Include in Phase 0

Add the EU AI Act readiness assessment to Phase 0 scope. It runs in parallel with the process audit and server setup — no additional elapsed time. Deliverable: a compliance register covering all six components above, completed before the build starts.

Cost: €6,000 – €9,000 (addition to Phase 0 fixed fee)

Timeline: 2–3 weeks, parallel with Phase 0

The key ongoing governance task is model version tracking: whenever the local Ollama model is updated, the AI inventory must be updated and the quarterly accuracy review must confirm metrics remain within acceptable bounds. This is a 2-hour quarterly task, not a programme.

Enterprise shippers that Transiberia wants to grow into increasingly ask logistics partners about AI governance. A completed assessment removes a commercial blocker before the first enterprise request for proposal arrives.

What IS required

- ✓ Carrier disclosure on WhatsApp bot
- ✓ Human confirmation before every action
- ✓ GDPR data retention policy for local database
- ✓ Quarterly AI accuracy monitoring log
- ✓ AI inventory updated on any model change
- ✓ Access controls on local server

What is NOT required

- ✗ Conformity assessment or notified body
- ✗ Algorithmic bias audit
- ✗ EU AI Act database registration
- ✗ Technical documentation per Annex IV
- ✗ Third-party AI provider audit
- ✗ Cross-border data transfer safeguards